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# =====
# Retrieval-Augmented Generation (RAG) over PDF
# LLM: TinyLlama/TinyLlama-1.1B-Chat-v1.0
# Google Colab Ready
# =====

# ----- INSTALL DEPENDENCIES -----
!pip install -q faiss-cpu transformers accelerate sentencepiece PyPDF2

# ----- IMPORTS -----
import faiss
import numpy as np
import torch
from transformers import AutoTokenizer, AutoModel,
AutoModelForCausalLM
from PyPDF2 import PdfReader
from google.colab import files

# ----- DEVICE -----
device = "cuda" if torch.cuda.is_available() else "cpu"
print("Using device:", device)

# ----- LOAD SENTENCE EMBEDDING MODEL -----
enc_name = "sentence-transformers/all-MiniLM-L6-v2"

enc_tokenizer = AutoTokenizer.from_pretrained(enc_name)
enc_model = AutoModel.from_pretrained(enc_name).to(device)
enc_model.eval()

# ----- LOAD LLM -----
llm_name = "TinyLlama/TinyLlama-1.1B-Chat-v1.0"

llm_tokenizer = AutoTokenizer.from_pretrained(llm_name)
llm_tokenizer.pad_token = llm_tokenizer.eos_token

llm_model = AutoModelForCausalLM.from_pretrained(
    llm_name,
    torch_dtype=torch.float16 if device=="cuda" else torch.float32,
    device_map="auto"
)

llm_model.eval()

# ----- UPLOAD PDF -----
uploaded = files.upload()
pdf_path = list(uploaded.keys())[0]

# ----- PDF TEXT EXTRACTION -----
def extract_text_from_pdf(path):
    reader = PdfReader(path)

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text = ""
for page in reader.pages:
    page_text = page.extract_text()
    if page_text:
        text += page_text + "\n"
return text

# ----- TEXT CHUNKING -----
def chunk_text(text, chunk_size=200, overlap=50):
    words = text.split()
    chunks = []

    start = 0
    while start < len(words):
        end = start + chunk_size
        chunks.append(" ".join(words[start:end]))
        start += chunk_size - overlap

    return chunks

# ----- EMBEDDING GENERATION -----
def encode_chunks(chunks, batch_size=8):
    all_embeddings = []

    with torch.no_grad():
        for i in range(0, len(chunks), batch_size):
            batch = chunks[i:i+batch_size]

            tokens = enc_tokenizer(
                batch,
                padding=True,
                truncation=True,
                return_tensors="pt"
            ).to(device)

            outputs = enc_model(**tokens)

            embeddings = outputs.last_hidden_state.mean(dim=1)

            all_embeddings.append(
                embeddings.cpu().numpy().astype("float32")
            )

    return np.vstack(all_embeddings)

```

```

# ----- BUILD VECTOR DATABASE -----
raw_text = extract_text_from_pdf(pdf_path)

doc_chunks = chunk_text(raw_text)

doc_embeddings = encode_chunks(doc_chunks)

faiss.normalize_L2(doc_embeddings)

dim = doc_embeddings.shape[1]

index = faiss.IndexFlatIP(dim)

index.add(doc_embeddings)

print("Total chunks indexed:", len(doc_chunks))

# ----- RETRIEVE CONTEXT -----
def retrieve_context(query, top_k=5):

    with torch.no_grad():

        tokens = enc_tokenizer(
            [query],
            padding=True,
            truncation=True,
            return_tensors="pt"
        ).to(device)

        outputs = enc_model(**tokens)

        q_emb = outputs.last_hidden_state.mean(dim=1)

        q_emb = q_emb.cpu().numpy().astype("float32")

        faiss.normalize_L2(q_emb)

        scores, indices = index.search(q_emb, top_k)

        return [doc_chunks[i] for i in indices[0]]

# ----- GENERATE RAG ANSWER -----
def generate_rag_answer(question, top_k=5, max_new_tokens=256):

    contexts = retrieve_context(question, top_k)

    context_text = "\n\n".join(contexts)

    prompt = f"""

```

You are an assistant that answers ONLY using the provided context.

Context:

{context_text}

Question: {question}

Answer:

"""

```
inputs = llm_tokenizer(
    prompt,
    return_tensors="pt",
    truncation=True
).to(device)

with torch.no_grad():

    output_ids = llm_model.generate(
        **inputs,
        max_new_tokens=max_new_tokens,
        do_sample=False
    )

    output_text = llm_tokenizer.decode(
        output_ids[0],
        skip_special_tokens=True
    )

    return output_text.split("Answer: ")[-1].strip()
```

----- RUN QUERY -----

question = "What is the main contribution of this paper?"

answer = generate_rag_answer(question)

print("\nQuestion:", question)

print("\nAnswer:", answer)

Using device: cpu

```
{"model_id": "a4a06a8925a54f5d85f0bcaa9b9a4a06", "version_major": 2, "version_minor": 0}
```

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2

Key	Status	
embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

```
{"model_id": "4dd9e840741f4491ac104862cde30bc9", "version_major": 2, "version_minor": 0}
```

WARNING:accelerate.big_modeling:Some parameters are on the meta device because they were offloaded to the disk and cpu.

<IPython.core.display.HTML object>

Saving w27392.pdf to w27392.pdf
Total chunks indexed: 65

This is a friendly reminder - the current text generation call has exceeded the model's predefined maximum length (2048). Depending on the model, you may observe exceptions, performance degradation, or nothing at all.

Question: What is the main contribution of this paper?

Answer: You are an assistant that answers ONLY using the provided context.

Context:

0.01 0.08 -5.00 -2.00 Time on social media d0.62*** (0.61) 0.69 0.24
0.00 1.00 Time news and online browsing d0.71*** (0.53) 0.75 0.21 1.00
1.00 Time online entertainment d0.74*** (0.54) 0.78 0.17 1.00 1.00 Time
in sports and exercised -0.46*** (0.75) 0.15 0.23 -1.00 0.00 Time
commuting d-0.89*** (0.36) 0.02 0.07 -1.00 -1.00 Time sleeping d0.17***
(0.83) 0.44 0.28 -1.00 1.00 a With and without COVID-19 levels are in
dollars and = percentage points difference. b With and without COVID-19
levels are in thousands of dollars and = percentage points difference.
c How the importance of this reason for choosing a major change due to
COVID-19. -1: decreased, 0: stayed the same, 1: increased. d How the
time allocated to each activity changed due to COVID-19. -1:
decreased, 0: stayed the same, 1: increased. Notes: : change. Prop.
>0: proportion of students for whom the individual level is positive.
Prop. = 0: proportion of students for whom the individual level is
zero. 25th and 75th percentiles of the cross-sectional distribution of
. Standard deviation in parentheses. *Significant at 10%, **5%, ***1%.
23 Table A2: Composition of COVID Effects: More Outcomes Expect earn at
age 35 (pp) Res wage (pp) Sem

Economic Proxies 0.000 0.002 0.002 0.031 0.116 0.166 0.001 0.003
Health Proxies 0.000 0.000 0.000 0.000 0.001 0.002 0.006 0.022
Controls Major FE Y Cohort FE Y
Y Mean 12.93 12.93 12.93 12.93
12.93 12.24 12.24 12.24 12.24 12.24 -4.18 -4.18 -4.18 -4.18 -4.18 -
13.39 -13.39 -13.39 -13.39 -13.39 R2 0.020 0.163 0.164 0.178 0.172

0.012 0.194 0.198 0.206 0.199 0.021 0.153 0.157 0.160 0.152 0.001
0.237 0.230 0.243 0.237 N 1446 1446 1446 1446 1446 1446 1446 1446 1446
1446 1446 1446 1446 1446 1446 1380 1380 1380 1380 1380 Notes: Standard
errors in parentheses bootstrapped with 1,000 replications. Each
column reports results from a separate OLS regression of the dependent
variable onto the covariates (row variables). Dependent variables
measured in percentage points. (:p <0:1,:p <0:05,:p <0:01) 20 Appendix
Figure A1: More Treatment Effects by Demographic Group (a)Withdrew from
Class due to COVID (0/1) (b)Social Events per Week (0-14) (c)Move in
With Family due to COVID (0/1) (d)Weekly Study Hours (

show that the negative economic and health impacts of COVID-19 have
been significantly more pronounced for less advantaged groups, and that
these are partially responsible for the underlying heterogeneity in
the impacts that we document. Our results suggest that by focusing on
addressing the economic and health burden imposed by COVID-19, as
measured by a relatively narrow set of mitigating factors, policy
makers may be able to prevent COVID-19 from widening existing
achievement gaps in higher education. 14 References Angrist, J. D. and
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Deitrick and Adam Streff provided excellent research assistance. All
errors that remain are ours. The views expressed herein are those of
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Student Experiences and Expectations: Evidence from a Survey Esteban

M. Aucejo, Jacob F. French, Maria Paola Ugalde Araya, and Basit Zafar
NBER Working Paper No. 27392 June 2020 JEL No. I2,I23,I24 ABSTRACT In

groups, and that could partly explain the heterogeneous treatment effects we documented in the previous section. 5.1 Economic and Health Mediating Factors We proxy for the financial and health shocks due to COVID-19 by relying on a small but relevant set of exogenous variables. We use the same set of variables as in the previous section, but with the addition of a new variable that captures the impact of the pandemic on the economy. We also include a new variable that captures the impact of the pandemic on health, which we define as the percentage of students who reported that they had been diagnosed with COVID-19 during the past 12 months. We use a logit model to estimate the impact of these variables on the dependent variable. We also estimate a model that includes theoretical covariates that we use to control for the potential confounding effects of the pandemic on the dependent variable. Weighted standard errors in parentheses. **5.2 Health Proxies 0.000 0.000 0.00 0.000 Health Proxies for 0:00 1:00 0:0: 1 0:00:1:in dollars and 0:0: 0: 1:00:831:00:80:0:00:

**0:in:00:in:
0: 0: